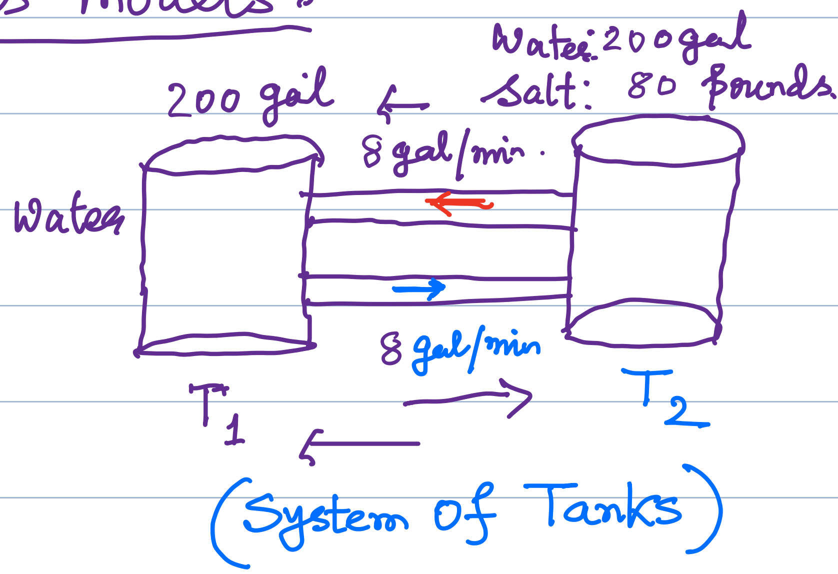
MTH 204: Lecture 12

System of ODEs

- **System of ODEs as models:**

(1) Mixing Tanks:



Problem: To determine the amount of salt in T_1 and T_2 at time t.

Let the amount of salt in T_1 at time t be $y_1(t)$ and the amount of salt in T_2 at time t be $y_2(t)$

$$y_1' = \text{inflow} - \text{outflow} = \left(\frac{y_2}{200} \times 8 - \frac{y_1}{200} \times 8\right) \frac{lb}{\cancel{gal}} \frac{\cancel{gal}}{min} = \frac{lb}{min}$$

and
$$y_2' = \text{inflow} - \text{outflow} = \left(\frac{y_1}{200} \times 8 - \frac{y_2}{200} \times 8\right) \frac{lb}{\cancel{gal}} \frac{\cancel{gal}}{min} = \frac{lb}{min}$$

$$\Rightarrow \left.\begin{array}{l} y_1' = -.04y_1 + .04y_2 \\ y_2' = .04y_1 - .04y_2 \end{array}\right\}$$

$$\Rightarrow \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} -.04 & .04 \\ .04 & -.04 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}.$$

$$Y = Y(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}$$

$$\Rightarrow \boxed{Y' = AY} \quad \text{where } A = \begin{pmatrix} -.04 & .04 \\ .04 & -.04 \end{pmatrix}$$

- We try a solution of the form: $Y = xe^{\lambda t}$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} e^{\lambda t}$$

$$\Rightarrow \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} \lambda x_1 e^{\lambda t} \\ \lambda x_2 e^{\lambda t} \end{pmatrix}$$

$$\Rightarrow Y' = \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \lambda e^{\lambda t} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Then $Y' = \lambda X e^{\lambda t} \Rightarrow \lambda x e^{\lambda t} = A x e^{\lambda t}$

$$\Rightarrow AX = \lambda X$$

Thus $Y = xe^{\lambda t}$ is a solution of the system of the ODE if X is an eigenvector of the matrix A corresponding to the eigenvalue λ of A.

Now $A = \begin{pmatrix} -.04 & .04 \\ .04 & -.04 \end{pmatrix}$

$$\det(A - \lambda I) = 0 \Rightarrow (\lambda + .04)^2 - (.04)^2 = 0$$

$$\Rightarrow \lambda^2 + .08\lambda = 0 \Rightarrow \lambda(\lambda + .08) = 0$$

$$\Rightarrow \lambda = 0, -.08$$

For $\lambda = 0$, $AX = 0.X \Rightarrow \begin{pmatrix} -.04 & .04 \\ .04 & -.04 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\left.\begin{array}{l} \Rightarrow -.04x_1 + .04x_2 = 0 \\ \Rightarrow x_1 = x_2 \end{array}\right\} \Rightarrow X_1 = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \left(\text{or } (1 \;\; 1)^T \right)$$

For $\lambda = -.08$, $AX = -.08X \Rightarrow \begin{pmatrix} -.04 & .04 \\ .04 & -.04 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -.08 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$$\left.\Rightarrow .04x_1 + .04x_2 = 0 \Rightarrow x_1 = -x_2 \right\} \Rightarrow X_2 = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\left(\text{or } (1 \; -1)^T \right)$$

Hence, the general solution is:

$$Y = c_1 X_1 e^{\lambda_1 t} + c_2 X_2 e^{\lambda_2 t}$$

$$= c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-.08t}$$

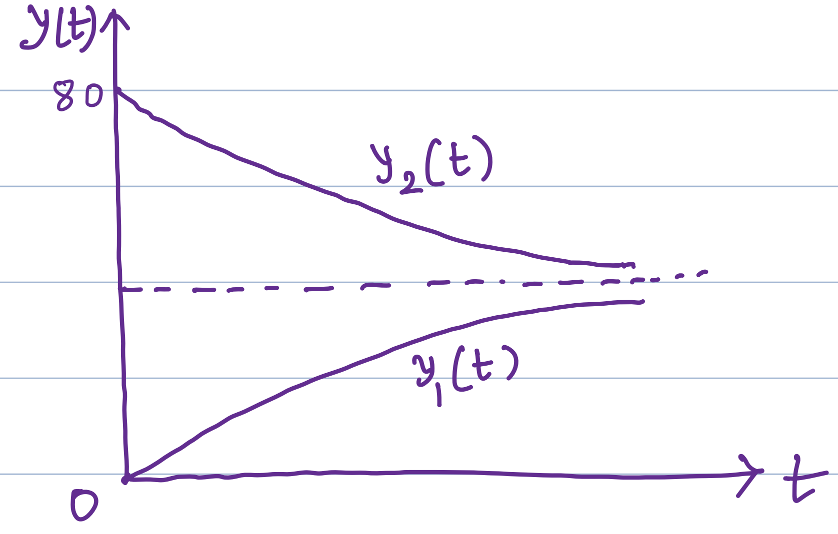
Now given $y_1(0) = 0$, $y_2(0) = 80$ $\left(\text{so, } Y(0) = \begin{pmatrix} 0 \\ 80 \end{pmatrix} \right)$

Then $c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 80 \end{pmatrix} \Rightarrow \left.\begin{array}{l} c_1 + c_2 = 0 \\ c_1 - c_2 = 80 \end{array}\right\}$

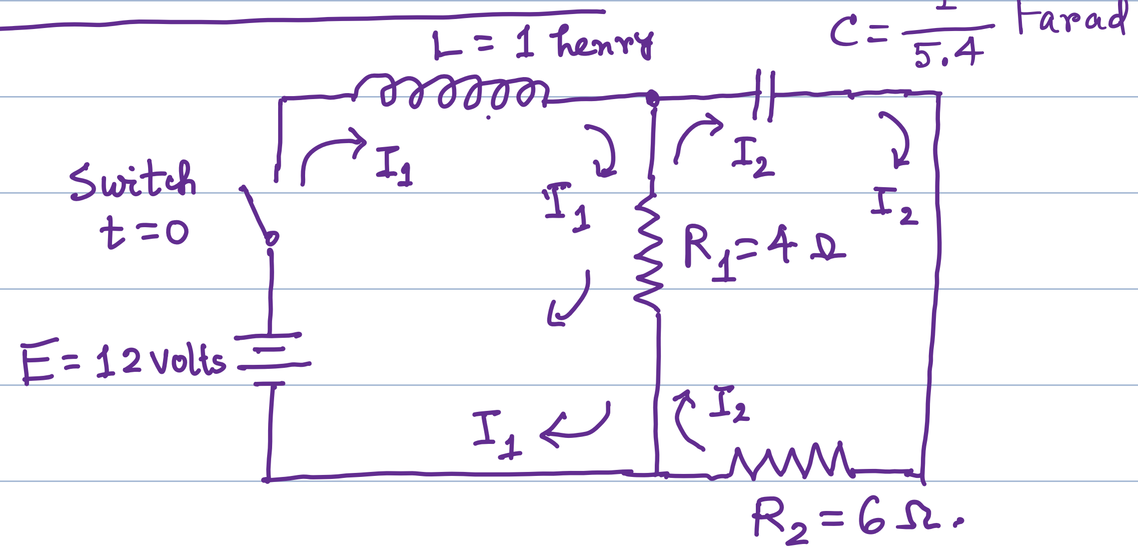
$$\Rightarrow c_1 = 40, \quad c_2 = -40$$

Therefore $Y = 40 \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 40 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-.08t}$

$$\Rightarrow \left.\begin{array}{l} y_1(t) = 40 - 40e^{-.08t} \\ y_2(t) = 40 + 40e^{-.08t} \end{array}\right\}$$



Ex: Electrical Network :



$$L I_1' + R_1 (I_1 - I_2) = E \quad \Rightarrow I_1' = -R_1 I_1 + R_1 I_2 + E$$

(given L = 1) .... (1)

and

$$R_1 (I_2 - I_1) + \frac{1}{C} \int I_2 \, dt + R_2 I_2 = 0$$

$$\Rightarrow R_1 (I_2' - I_1') + \frac{1}{C} I_2 + R_2 I_2' = 0$$

$$\Rightarrow (R_1 + R_2) I_2' - R_1 I_1' + \frac{1}{C} I_2 = 0$$

$$\Rightarrow (R_1+R_2)I_2' - R_1(-R_1 I_1 + R_1 I_2 + E) + \frac{1}{C} I_2 = 0 \quad \cdots\cdots (2)$$

Now $R_1 = 4$, $R_2 = 6$, $E = 12$, $C = \frac{1}{5.4}$

Then (1) $\Rightarrow$ $\boxed{I_1' = -4I_1 + 4I_2 + 12}$

and (2) $\Rightarrow$ $10 I_2' - 4(-4I_1 + 4I_2 + 12) + \frac{I_2}{\frac{1}{5.4}} = 0$

$$\Rightarrow I_2' = .4(-4I_1 + 4I_2 + 12) - \frac{5.4 I_2}{10} = 0$$

$$\Rightarrow I_2' = -1.6 I_1 + 1.6 I_2 + 4.8 - .54 I_2$$

$$\Rightarrow \boxed{I_2' = -1.6 I_1 + 1.06 I_2 + 4.8}$$

Thus the system of ODE is

$$\left.\begin{aligned} I_1' &= -4I_1 + 4I_2 + 12 \\ I_2' &= -1.6 I_1 + 1.06 I_2 + 4.8 \end{aligned}\right\}$$

or $$\begin{pmatrix} I_1' \\ I_2' \end{pmatrix} = \begin{pmatrix} -4 & 4 \\ -1.6 & 1.06 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} + \begin{pmatrix} 12 \\ 4.8 \end{pmatrix}$$

This is a nonhomogeneous system of ODE

First we find the general solution of the homogeneous system of ODE

$$I' = AI \quad \text{where } I = \begin{pmatrix} I_1 \\ I_2 \end{pmatrix}$$

and $A = \begin{pmatrix} -4 & 4 \\ -1.6 & 1.06 \end{pmatrix}$

First we find the eigen values of A.

$\det(A - \lambda I) = 0 \Rightarrow (-4-\lambda)(1.06-\lambda) - 4(-1.6) = 0$

$\Rightarrow -4.24 + 2.94\lambda + \lambda^2 + 6.4 = 0$

$\Rightarrow \lambda^2 + 2.94\lambda + 2.16 = 0$

$\Rightarrow (\lambda + 1.5)(\lambda + 1.44) = 0$

So, the eigen values are $\lambda = -1.5, -1.44$

For eigen vector corresponding to the eigenvalue $\lambda = -2$

we solve: $\begin{pmatrix} -4 & 4 \\ -1.6 & 1.06 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -1.5 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$\Rightarrow \left.\begin{array}{l} -2.5x_1 + 4x_2 = 0 \\ -1.6x_1 + 2.56x_2 = 0 \end{array}\right\}$

$\Rightarrow x_2 = \frac{2.5}{4} x_1 = .625 x_1$

So, $\begin{pmatrix} 1 \\ .625 \end{pmatrix}$ is an eigen vector corresponding to the eigen value $\lambda = -1.5$

For eigen vector corresponding to the eigen value $\lambda = -.8$

we solve:
$$\begin{pmatrix} -4 & 4 \\ -1.6 & 1.06 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -1.44 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\Rightarrow \left.\begin{aligned} -2.56x_1 + 4x_2 &= 0 \\ -1.6x_1 + 2.5x_2 &= 0 \end{aligned}\right\}$$

$$\Rightarrow x_2 = \frac{2.56}{4} x_1 = .64 x_1$$

$\begin{pmatrix} 1 \\ .64 \end{pmatrix}$ is an eigen vector corresponding to the eigen value $\lambda = -1.44$

So, the general solution of the homogeneous system of ODE is:

$$I = \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ .625 \end{pmatrix} e^{-1.5t} + c_2 \begin{pmatrix} 1 \\ .64 \end{pmatrix} e^{-1.44t}$$

Now for a particular solution of the nonhomogeneous system of ODE $I' = AI + C$ $\left[\text{Here } C = \begin{pmatrix} 12 \\ 4.8 \end{pmatrix}\right]$

we try a constant vector $I = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$.

Then
$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 & 4 \\ -1.6 & 1.06 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} 12 \\ 4.8 \end{pmatrix}$$

$$-4a_1 + 4a_2 + 12 = 0$$
$$-1.6a_1 + 1.06a_2 + 4.8 = 0$$

$$\left.\begin{array}{l} 4a_1 - 4a_2 = 12 \Rightarrow a_1 - a_2 = 3 \\ -1.6a_1 + 1.06a_2 = 4.8 \Rightarrow 1.6a_1 - 1.06a_2 = 4.8 \end{array}\right\}$$

$$1.6a_1 - 1.6a_2 = 4.8$$
$$1.6a_1 - 1.06a_2 = 4.8$$

Subtracting, $-.54a_2 = 0 \Rightarrow a_2 = 0$

Now from $a_1 - a_2 = 3$, we get $a_1 - 0 = 3$
$$\Rightarrow a_1 = 3$$

So, the general solution of the nonhomogeneous ODE is :
$$I = \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ .625 \end{pmatrix} e^{-1.5t} + c_2 \begin{pmatrix} 1 \\ .64 \end{pmatrix} e^{-1.44t} + \begin{pmatrix} 3 \\ 0 \end{pmatrix}$$

where c_1 and c_2 are arbitrary constants.

To determine c_1, c_2 we use the initial condition $I(0) = 0$

Then $c_1 + c_2 = -3 \Rightarrow .625c_1 + .625c_2 = -1.875$

and $.625c_1 + .64c_2 = 0 \Rightarrow -.625c_1 + .64c_2 = 0$

$\Rightarrow -.015c_2 = -1.875$

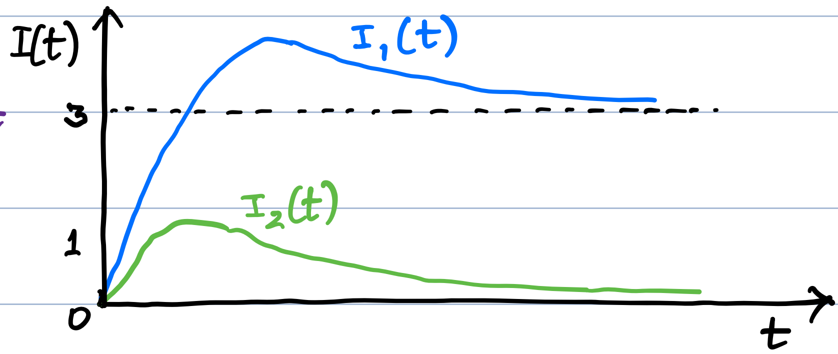
$\Rightarrow c_2 = \frac{1.875}{.015} = \boxed{125}$

Then $c_1 = -3 - 125 = \boxed{-128}$

Thus the solution of the nonhomogeneous system of ODE is: $I = -128\begin{pmatrix}1\\.625\end{pmatrix}e^{-1.5t} + 125\begin{pmatrix}1\\.64\end{pmatrix}e^{-1.44t} + \begin{pmatrix}3\\0\end{pmatrix}$

$\Rightarrow I_1(t) = -128e^{-1.5t} + 125e^{-1.44t} + 3$

$I_2(t) = -80e^{-1.5t} + 80e^{-1.44t}$



Conversion of an n-th order ODE to a system of ODE:

An n-th order ODE $y^{(n)} = F(t, y, y', \ldots, y^{(n-1)})$ can be converted to a system of n first order ODEs by setting $y = y_1$ and

$$\begin{aligned} y_1' &= y_2 \\ y_2' &= y_3 \\ &\vdots \\ y_{n-1}' &= y_n \\ y_n' &= F(t, y_1, y_2, \ldots, y_n) \end{aligned}$$

Note that $y_1 = y$, $y_2 = y_1' = y'$, $y_3 = y_2' = y''$, ... $y_{n-1} = y_{n-2}' = y^{(n-2)}$,
$y_n = y_{n-1}' = y^{(n-1)}$

Ex: Consider the second order ODE :

$$my'' + cy' + ky = 0$$

Then $y'' = -\frac{c}{m}y' - \frac{k}{m}y$

Let $y = y_1$, $y_1' = y_2$ So, $y' = y_2$ and $y'' = y_2'$

Therefore the second order ODE can be written as a system of ODE :

$$\left.\begin{aligned} y_1' &= y_2 \\ y_2' &= -\frac{k}{m}y_1 - \frac{c}{m}y_2 \end{aligned}\right\}$$

$$\Rightarrow \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

The characteristic equation is :

$$\begin{vmatrix} -\lambda & 1 \\ -\frac{k}{m} & -\frac{c}{m} - \lambda \end{vmatrix} = 0$$

$$\Rightarrow -\lambda\left(-\frac{c}{m} - \lambda\right) - \left(-\frac{k}{m}\right) = 0$$

$$\Rightarrow \lambda^2 + \frac{c}{m}\lambda + \frac{k}{m} = 0$$

Let us now assume that in this problem,

$m=2$, $c=8$ and $k=6$

Then the characteristic equation becomes:

$$\lambda^2 + 4\lambda + 3 = 0$$

$$\Rightarrow \lambda^2 + 3\lambda + \lambda + 3 = 0$$

$$\Rightarrow \lambda(\lambda+3) + (\lambda+3) = 0$$

$$\Rightarrow (\lambda+3)(\lambda+1) = 0$$

So, the two eigen values are $\lambda = -1$, $\lambda = -3$

For eigen vector corresponding to the eigenvalue $\lambda = -1$ we solve:

$$\begin{pmatrix} 0 & 1 \\ -3 & -4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (-1)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\Rightarrow \left.\begin{aligned} x_2 + x_1 &= 0 \\ -3x_1 - 3x_2 &= 0 \end{aligned}\right\} \Rightarrow x_2 = -x_1$$

Thus $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an eigen vector corresponding to the eigen value $\lambda = -1$

For eigen vector corresponding to the eigenvalue $\lambda = -3$, we solve:

$$\begin{pmatrix} 0 & 1 \\ -3 & -4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (-3)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\Rightarrow \left.\begin{array}{l} 3x_1 + x_2 = 0 \\ -3x_1 - x_2 = 0 \end{array}\right\} \Rightarrow x_2 = -3x_1$$

Hence $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$ is an eigen vector corresponding to the eigen value $\lambda = -3$

Thus the general solution of the system of ODE is:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ -3 \end{pmatrix} e^{-3t}$$

Note that the first component is

$$y = y_1 = c_1 e^{-t} + c_2 e^{-3t}$$

(which is the expected solution of the second order ODE)

and the second component is y_2 which is the derivative of the first component y_1.